

PARA – Polyarylamide PARA CF40

AKROLOY® PARA ICF 40 black (6128)

Tensile modulus

39000 MPa

1 mm/min

ISO 527-2

Stress at break

280 MPa

5 mm/min

ISO 527-2

Charpy impact strength

35 kJ/m²

23°C

ISO 179-1/1eU

AKROLOY® PARA ICF 40 black (6128) is a 40% carbon fibre reinforced polyarylamid with good flexural strength and good tribological properties.

Typical applications

Components with high dimensional stability, independent from moisture content.



Mechanical Properties

Tensile modulus (1 mm/min | ISO 527-2)

d.a.m.

39000 MPa

conditioned

38000 MPa

Stress at break (5 mm/min | ISO 527-2)

d.a.m.

280 MPa

conditioned

255 MPa

Strain at break (5 mm/min | ISO 527-2)

d.a.m.

1,0 %

conditioned

1,0 %

Flexural modulus (2 mm/min | ISO 178)

d.a.m.

37000 MPa

conditioned

35000 MPa

Flexural strength (2 mm/min | ISO 178)

d.a.m.

420 MPa

conditioned

360 MPa

Flexural strain at break (2 mm/min | ISO 178)

d.a.m.

1,2 %

conditioned

1,1 %

Charpy impact strength (23°C | ISO 179-1/1eU)

d.a.m.

35 kJ/m²

conditioned

35 kJ/m²

Charpy notched impact strength (23°C | ISO 179-1/1eA)

d.a.m.

5 kJ/m²

conditioned

5 kJ/m²



Thermal Properties

Temperature of deflection under load HDT/A (1,8 MPa | ISO 75)

230 °C

Melting temperature (DSC, 10K/min | DIN EN ISO 11357-3)

235 °C



Flammability

Burning rate (UL 94)

1,6mm Wall thickness

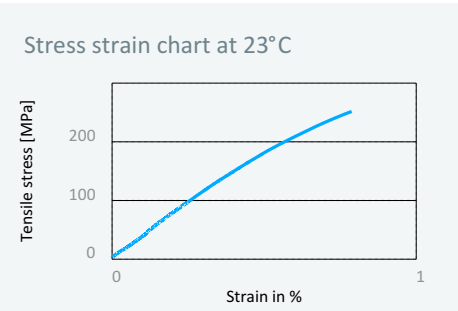
HB Class

Burning rate (<100 mm/min) (> 1 mm Thickness | FMVSS 302)

+

Disclaimer:

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General properties

Density (23°C ISO 1183)	1,4 g/cm ³
Humidity absorption (70°C, 62% r.H. ISO 1110)	1,02 %
Molding shrinkage (flow ISO 294-4)	0,1 - 0,2 %
Molding shrinkage (transverse ISO 294-4)	0,2 - 0,4 %



Electrical Properties

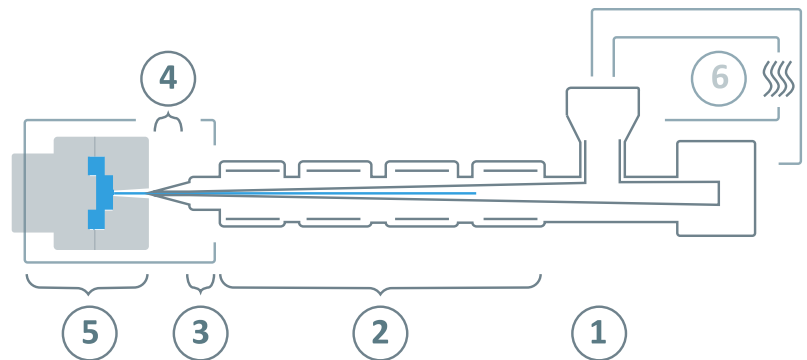
Surface resistivity (acc. to IEC 60093)	
d.a.m.	1,0E+4 Ohm
conditioned	1,0E+4 Ohm

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Processing information

The listed values are recommendations. Higher values should be used for higher glass loadings. We recommend only dehumidifying or vacuum dryers. Extensive drying can cause filling problems and surface defects.



⑥	Drying time	4 - 12 h
	Drying temperature ($\tau \leq -30^{\circ}\text{C}$)	80 - 90°C
	Processing moisture	0,02 - 0,1%
①	Feed section	60 - 80°C
②	Temperature zone 1 - Zone 4	250 - 300°C
③	Nozzle temperature	270 - 300°C
④	Melt temperature	270 - 300°C
⑤	Mold temperature	120 - 160°C
→	Holding pressure, spec.	300 - 1500 bar
←	Back pressure, spec.	50 - 150 bar
	Injection speed	high
	Screw speed	8 - 10 m/min

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